SUPPORTING TRANSMISSION GRID VOLTAGE CONTROL WITH ACTIVE DISTRIBUTION GRIDS USING ONLINE FEEDBACK OPTIMISATION

Manuel Schwenke[1], Felix Korff[1], Jutta Hanson[1]*

[1]*Department of Electrical Engineering and Information Technology,*
Institute of Electrical Power Supply with Integration of Renewable Energy (E5),
Technical University of Darmstadt, Landgraf-Georg-Straße 4, 64283 Darmstadt, Germany
** manuel.schwenke@tu-darmstadt.de*

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Abstract

The transition to decentralised electricity generation enables the integration of active distribution grids into transmission grid voltage control as an ancillary service. This study presents a mixed-integer online feedback reactive power controller. The controller iteratively optimises reactive power provision of distribution grids, minimising losses in the distribution grid while ensuring adherence to operational limits and voltage setpoints at interconnection nodes. The controller's functionality is demonstrated using a high-voltage benchmark grid. The controller exhibits robust performance without requiring detailed knowledge of the distribution or transmission grid state. The results highlight the efficiency of online feedback optimisation in reactive power supply with minimal losses, supporting the economic and reliable provision of ancillary services.

1 Introduction

The increasing integration of renewable energy resources into the electrical grid necessitates innovative approaches to maintain stability and efficiency. Among the critical challenges is the provision of reactive power from distribution grids (DG) as an ancillary service for transmission grid (TG) operation. These novel possibilities are also outlined in regulatory frameworks, e.g. the system stability roadmap of the Federal Ministry for Economic Affairs and Climate Protection in Germany, which demands a systematic integration of DG into the provision of reactive power for the transmission grid to ensure voltage stability and system reliability [1].

The provision of reactive power has traditionally relied on resources at the transmission level. In grids characterised by distributed renewables, where the share of synchronous generators is decreasing, these resources are increasingly scarce. Harnessing the flexibility of DG is a potential way to support the TG with reactive power, making better use of existing resources and potentially leading to lower investment costs for compensation equipment in the TG [2]. This introduces unique challenges, such as avoiding additional power losses and maintaining voltage stability within the DG itself, while providing reactive power at the interconnection to the TG. In [3], voltage support schemes at the TG-DG interconnection are reviewed across different countries. So far hardly any Transmission System Operator (TSO) uses automated concepts for voltage support at the TG-DG interconnection. Instead, TSO often rely on manual communication methods, such as telephone calls, to request reactive power from Distribution System Operator (DSO) during operational planning.

In regular operation, the secondary side of TG-DG transformers is controlled to ensure a safe supply voltage in the DG. The proposed approach leverages a feedback controller to control the primary side according to the specifications of the TSO. The TSO designs its specifications in such a way that voltage maintenance and static voltage stability in the TG is ensured. Therefore the TSO defines bounds or a specific setpoint for the voltage magnitudes at the primary side of the interconnection transformer, e.g. in the form of a voltage schedule for operational planning, so that reactive power reserve and the distance to voltage collapse points for remaining PQ load buses is increased. The DSO utilises a controller to manage resources and to ensure that the voltage requirements are met.

In this context, online feedback optimisation (OFO) emerges as a promising methodology. Unlike conventional optimisation techniques, OFO sequentially solves simple quadratic programs to compute a gradient for the input signals, minimising an objective while ensuring input and output signals remain within feasible bounds [4]. The output signals are fed back to the controller, creating a feedback loop. The feedback nature offers advantages such as robustness against data uncertainties, including erroneous forecasts or imprecise measurements

The controller presented in Sec. 2 uses both real-valued control variables, namely the reactive power feed-in of distributed energy resources (DER) and integer-valued control variables, namely the transformer tap positions. The controller's functionality is demonstrated in Sec. 3 and Sec. 4 using a high-voltage grid of the Simbench data set. Sec. 5 summarises the findings and provides a basis for further research.

2 Mixed-Integer Online Feedback Optimisation

2.1 Optimisation Model / Controller

Based on [5], the online feedback optimisation (OFO) is realised with a time discrete feedback controller that updates the input vector $\boldsymbol{u}$ iteratively:

$$\boldsymbol{u}_{k+1} = \boldsymbol{u}_k + \alpha \boldsymbol{\sigma}(\boldsymbol{u}_k, \boldsymbol{d}_k, \boldsymbol{y}_k), \tag{1}$$

where $\boldsymbol{\sigma}(\boldsymbol{u}_k, \boldsymbol{d}_k, \boldsymbol{y}_k)$ is the solution of the following Mixed-Integer Quadratic Program (MIQP):

$$\boldsymbol{\sigma}_k = \arg\min_{\boldsymbol{w}_k} \|\boldsymbol{w}_k + \mathbf{G}_\mathbf{w}^{-1}\boldsymbol{\Gamma}(\boldsymbol{u}_k, \boldsymbol{d}_k)^T \boldsymbol{\nabla} \boldsymbol{f}(\boldsymbol{u}_k, \boldsymbol{y}_k)\|^2_{\mathbf{G}_\mathbf{w}}, \tag{2}$$

$$\text{s.t. } \mathbf{A}_\mathrm{u}(\boldsymbol{u}_k + \alpha \boldsymbol{w}_k) \leq \boldsymbol{b}_{\mathrm{u},k}, \tag{3}$$

$$\mathbf{A}_\mathrm{y}(\boldsymbol{y}_k + \alpha \boldsymbol{\nabla}_{\boldsymbol{u}} \mathbf{H}(\boldsymbol{u}_k, \boldsymbol{d}_k)\boldsymbol{w}_k) \leq \boldsymbol{b}_{\mathrm{y},k}, \tag{4}$$

$$\boldsymbol{w}_k = [\boldsymbol{w}_{\mathbb{R},k}, \boldsymbol{w}_{\mathbb{Z},k}], \boldsymbol{w}_{\mathbb{R},k} \subset \mathbb{R}, \boldsymbol{w}_{\mathbb{Z},k} \subset \mathbb{Z}. \tag{5}$$

The controller sequentially solves (2) to determine $\boldsymbol{w}_k$ and updates the input variables $\boldsymbol{u}_{k+1}$ according to (1). In (2), $\boldsymbol{\Gamma}(\boldsymbol{u}_k, \boldsymbol{d}_k)^T$ is defined as $[\mathbf{I}_{n_\mathrm{u}} \ \mathbf{H}(\boldsymbol{u}_k, \boldsymbol{d}_k)]^T$, where $\mathbf{I}_{n_\mathrm{u}}$ represents the identity matrix of dimension n_u, and $\mathbf{H}(\boldsymbol{u}_k, \boldsymbol{d}_k)$ denotes the steady-state input-output sensitivities. Control disturbances are represented with $\boldsymbol{d}$, i.e. active power feed-in or loads. The vector $\boldsymbol{\nabla f}$ is the control objective. $\mathbf{A}_\mathrm{u}$ and $\boldsymbol{b}_\mathrm{u}$ define linear input, $\mathbf{A}_\mathrm{y}$ and $\boldsymbol{b}_\mathrm{y}$ linear output constraints.

The controller is fed with measurements of the output variables $\boldsymbol{y}_k$ with a sampling rate of $\Delta t_\mathrm{m} = 1$ min. The sampling rate should be at least high enough for the system to be in a steady state at the time of sampling. The MIQP in (2) uses the fed back measurements to determine an optimal gradient $\boldsymbol{\sigma}_k = \boldsymbol{w}_k^*$ such that the norm is minimised using the open-source solver SCIP [6]. Mathematically, this means a projection of the gradient of the objective $\boldsymbol{\nabla f}$ onto the feasible definition space, i.e. $\boldsymbol{u}_k + \alpha \boldsymbol{w}_k \in \mathcal{U}_k, \boldsymbol{y}_k + \alpha \boldsymbol{\nabla}_{\boldsymbol{u}} \mathbf{H} \in \mathcal{Y}_k$. The controller input is updated sequentially according to (1) with the controller gain α and the positive-definite matrix $\mathbf{G}_\mathbf{w}$. The tuning of α and $\mathbf{G}_\mathbf{w}$ is discussed in Sec 2.5.

2.2 Input/Output Variables and Bounds

The input vector $\boldsymbol{u}_k = [\boldsymbol{u}_{\mathbb{R},k}, \boldsymbol{u}_{\mathbb{Z},k}]$ consists of real-valued reactive power setpoints for DER and integer-valued tap positions for the tap changers of the interconnection transformers. The feasible range of $\boldsymbol{u}_{\mathbb{R},k}$ for DER connected to HV grids is defined by the operating diagrams shown in Fig. 1. To take compliance into account, $\boldsymbol{u}_k$ must also be measured and fed back to the controller. The vector $\boldsymbol{u}_{\mathbb{Z},k}$ is a set of integers defined by the amount of mechanically settable tap changer positions. In the study case examined later, the constraint $-16 \leq \boldsymbol{u}_{\mathbb{Z},k} \leq 16$ applies, and tap steps can be adjusted in increments of $\Delta\tau = 1\%$ of the nominal transmission ratio. The output vector $\boldsymbol{y}_k$ consists of real-valued voltage magnitudes and is obtained by a Linearised Output Projection (LOP). A thorough introduction to LOP and the theory of projected gradient flows can be found in [4]. The linearised relationship

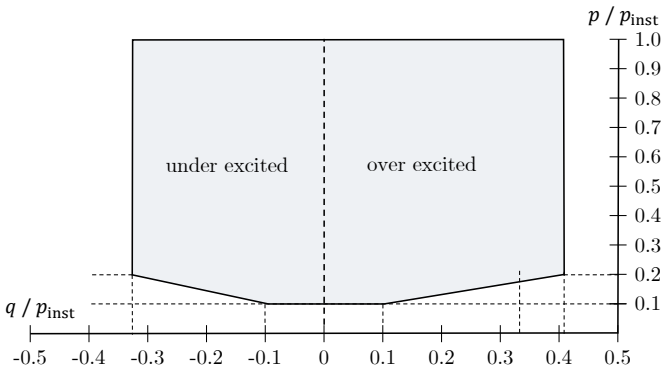

Fig. 1 Reactive power capability chart for DER connected to HV grids, according to VDE-AR-N 4120 [7].

between $\boldsymbol{u}_k$ and $\boldsymbol{y}_k$ is given by $\boldsymbol{\nabla}\mathbf{H}$. This makes the controller feasible with respect to $\mathcal{U}_k$ and approximately feasible with respect to $\mathcal{Y}_k$ up to the first order because of the inexact linearised input-output sensitivities $\boldsymbol{\nabla}\mathbf{H}$.

In contrast to [5], setpoint specifications for voltage magnitudes are to be implemented as soft constraints. The objective (2) is augmented with the term $\boldsymbol{s}_k \mathbf{G}_\mathrm{s} \boldsymbol{s}_k^T$, similar to the approach presented in [8]. This formulation ensures that the controller finds a solution even if the control reserve is insufficient. However, this allows violations of the voltage band. With the bounds for $\boldsymbol{u}_k$ and $\boldsymbol{y}_k$ and the simplification of (2) equations (2)-(4) are reformulated to obtain (6)-(10). Terms in (2) that do not depend on $\boldsymbol{w}_k$ can be neglected, as they do not influence the minimiser $\boldsymbol{w}_k^*$.

$$\boldsymbol{\sigma}_k = \min_{\boldsymbol{w}_k, \boldsymbol{s}_k} \boldsymbol{w}_k \mathbf{G}_\mathrm{w} \boldsymbol{w}_k^T + \boldsymbol{\nabla f} \boldsymbol{w}_k^T + \boldsymbol{s}_k \mathbf{G}_\mathrm{s} \boldsymbol{s}_k^T, \tag{6}$$

$$\text{s.t. } \alpha \mathbf{I}_{n_\mathrm{u}} \leq \boldsymbol{u}_{\max,k} - \boldsymbol{u}_{\mathrm{m},k}, \tag{7}$$

$$\alpha \mathbf{I}_{n_\mathrm{u}} \geq \boldsymbol{u}_{\min,k} - \boldsymbol{u}_{\mathrm{m},k}, \tag{8}$$

$$\alpha \boldsymbol{\nabla}_{\boldsymbol{u}} \mathbf{H} \boldsymbol{w}_k \leq \boldsymbol{y}_{\max,k} - \boldsymbol{y}_{\mathrm{m},k}, \tag{9}$$

$$\alpha \boldsymbol{\nabla}_{\boldsymbol{u}} \mathbf{H} \boldsymbol{w}_k \geq \boldsymbol{y}_{\min,k} - \boldsymbol{y}_{\mathrm{m},k}. \tag{10}$$

All measured variables have the index m. The bounds $\boldsymbol{y}_{\max,k}$ and $\boldsymbol{y}_{\min,k}$ can be defined by the voltage band, i.e 0.9 pu and 1.1 pu or a specific voltage setpoint, i.e $\boldsymbol{y}_{\max,k} = \boldsymbol{y}_{\min,k}$.

2.3 Sensitivities

The proposed OFO controller uses linearisations of the power flow equations. The actual non-linear model that results in time-varying sensitivities $\boldsymbol{\nabla}\mathbf{H}(\boldsymbol{u}_k, \boldsymbol{d}_k)$ does not need to be known due to the feedback nature of the controller. A constant sensitivity of the input to the output variables $\boldsymbol{\nabla}\mathbf{H} = \mathrm{const.}$ is used. This shows the particular strength of OFO, because model errors, forecast errors or even topology changes do not jeopardise compliance with technical constraints, as long as sufficient control reserve is available. However, if the controller is provided with the correct sensitivity because the grid is fully observable, the convergence behaviour improves. The sensitivity is determined here once according to (11) before the start of the simulation from the case of low load with maximum feed-in

of DER and then used for all time steps k:

$$\nabla \mathbf{H} = [\nabla_{\boldsymbol{u}_{\mathbb{R}}} \mathbf{H},\ \nabla_{\boldsymbol{u}_{\mathbb{Z}}} \mathbf{H}]\,, \tag{11}$$

with

$$\boldsymbol{\nabla}_{\boldsymbol{u}_{\mathbb{R}}} \mathbf{H} = \mathbf{F}_{\mathrm{QV}}^{-1} \mathbf{C}, \tag{12}$$

where $\mathbf{F}_{\mathrm{QV}}^{-1}$ is the inverse of element $\mathbf{F}_{\mathrm{QV}}$ of the Jacobian:

$$\mathbf{J} = \begin{bmatrix} \mathbf{F}_{\mathrm{P}\theta} & \mathbf{F}_{\mathrm{PV}} \\ \mathbf{F}_{\mathrm{Q}\theta} & \mathbf{F}_{\mathrm{QV}} \end{bmatrix} = \begin{bmatrix} \frac{\partial \boldsymbol{p}}{\partial \boldsymbol{\theta}} & \frac{\partial \boldsymbol{p}}{\partial |\boldsymbol{V}|} \\ \frac{\partial \boldsymbol{q}}{\partial \boldsymbol{\theta}} & \frac{\partial \boldsymbol{q}}{\partial |\boldsymbol{V}|} \end{bmatrix}, \tag{13}$$

and $\mathbf{C}$ is a $n_{\mathrm{b}} \times n_{\boldsymbol{u}_{\mathbb{R}}}$ connection matrix, with the amount of buses n_{b} and the amount of DER $n_{\boldsymbol{u}_{\mathbb{R}}}$, where in each column j all entries C_{ij} of $\mathbf{C}$ are 0, except for i corresponding to the index of the bus a DER is connected to, where it is 1. As the derivation of the sensitivity $\boldsymbol{\nabla}_{\boldsymbol{u}_{\mathbb{Z}}} \mathbf{H} = \partial |\underline{\boldsymbol{V}}| / \partial \boldsymbol{\tau}$ with the tap position τ is quite extensive, readers are referred to [9].

In addition to the possibility of updating the sensitivity in case of full observability due to sufficient measurements, possibly as a result of a state estimation, techniques of adaptive control can be used to estimate the time-varying sensitivity, e.g. by a Kalman filter, see [10].

2.4 Objective

OFO can be used to minimise an objective $\boldsymbol{f}(\boldsymbol{u}_k, \boldsymbol{d}_k, \boldsymbol{y}_k)$. The objective is a function of $\boldsymbol{u}_k$, $\boldsymbol{d}_k$ or $\boldsymbol{y}_k$ and minimising it leads to a desired operating state of the system. In this case, the objective is the total losses of the HV grid, i.e. $\boldsymbol{f}(\boldsymbol{u}_k, \boldsymbol{d}_k, \boldsymbol{y}_k) = \boldsymbol{p}_{\mathrm{l}}(\boldsymbol{u}_k, \boldsymbol{d}_k, \boldsymbol{y}_k)$. The controller does not need to know the functional relationship $\boldsymbol{p}_{\mathrm{l}}(\boldsymbol{u}_k, \boldsymbol{d}_k, \boldsymbol{y}_k)$, but only the gradient $\boldsymbol{\nabla} \boldsymbol{p}_{\mathrm{l}}$. Here, $\boldsymbol{\nabla} \boldsymbol{p}_{\mathrm{l}} = \mathrm{const.}$ is derived once from a system state with low load and maxium feed-in, exactly like $\boldsymbol{\nabla} \mathbf{H}$. The gradient of the objective with respect to $\boldsymbol{u}_{\mathbb{R}}$ can be calculated as follows:

$$\boldsymbol{\nabla}_{\boldsymbol{u}_{\mathbb{R}}} \boldsymbol{f} = \frac{\partial \boldsymbol{p}_{\mathrm{l}}}{\partial |\underline{\boldsymbol{v}}|} \frac{\partial |\underline{\boldsymbol{v}}|}{\partial \boldsymbol{q}_{\mathrm{DER}}} = \frac{\partial \boldsymbol{p}_{\mathrm{l}}}{\partial |\underline{\boldsymbol{v}}|} \boldsymbol{\nabla} \mathbf{H}, \tag{14}$$

with:

$$\begin{aligned} \frac{\partial \boldsymbol{p}_{\mathrm{l}}}{\partial |\underline{\boldsymbol{v}}|} = \mathrm{Re} \Big\{ & \operatorname{diag}(\underline{\boldsymbol{i}}_{\mathrm{f}})^* \mathbf{C}_{\mathrm{f}} \operatorname{diag}(|\underline{\boldsymbol{v}}|^{-1} \underline{\boldsymbol{v}})^* \\ & + \operatorname{diag}(\mathbf{C}_{\mathrm{f}} \underline{\boldsymbol{v}}) \mathbf{Y}_{\mathrm{f}}{}^* \operatorname{diag}(|\underline{\boldsymbol{v}}|^{-1} \underline{\boldsymbol{v}})^* \\ & - \operatorname{diag}(\underline{\boldsymbol{i}}_{\mathrm{t}})^* \mathbf{C}_{\mathrm{t}} \operatorname{diag}(|\underline{\boldsymbol{v}}|^{-1} \underline{\boldsymbol{v}})^* \\ & - \operatorname{diag}(\mathbf{C}_{\mathrm{t}} \underline{\boldsymbol{v}}) \mathbf{Y}_{\mathrm{t}}{}^* \operatorname{diag}(|\underline{\boldsymbol{v}}|^{-1} \underline{\boldsymbol{v}})^* \Big\}, \end{aligned} \tag{15}$$

where $\underline{\boldsymbol{v}}$ is the vector of complex nodal voltages, $\underline{\boldsymbol{i}}_{\mathrm{f}}$, $\underline{\boldsymbol{i}}_{\mathrm{t}}$ are the vectors of branch currents at the from and to end and $\mathbf{C}_{\mathrm{f}}$, $\mathbf{C}_{\mathrm{t}}$ are the branch connection matrices [11]. The gradient $\boldsymbol{\nabla} \mathbf{f}_{\boldsymbol{u}_{\mathbb{Z}}}$ of the objective with respect to $\boldsymbol{u}_{\mathbb{Z}}$ is:

$$\boldsymbol{\nabla}_{\boldsymbol{u}_{\mathbb{Z}}} \boldsymbol{f} = \frac{\partial \boldsymbol{p}_{\mathrm{l}}}{\partial |\underline{\boldsymbol{v}}|} \frac{\partial |\underline{\boldsymbol{v}}|}{\partial \boldsymbol{\tau}}. \tag{16}$$

The gradient of the objective is then composed as follows:

$$\nabla \boldsymbol{f} = [\nabla_{\boldsymbol{u}_{\mathbb{R}}} \boldsymbol{f},\ \nabla_{\boldsymbol{u}_{\mathbb{Z}}} \boldsymbol{f}]\,. \tag{17}$$

2.5 Controller Tuning

The coefficient α and the tuning matrix $\mathbf{G}_{\mathrm{w}}$ influence the convergence behaviour of the controller. An entry $\mathrm{G}_{\mathrm{w},i,j}$ also influences how strongly a control variable w_j is used to minimise the control objective $\boldsymbol{\nabla} \boldsymbol{f}$. The parameterisation is done experimentally and is $\alpha = 1$, $g_{\mathrm{q}} = 0.01$ and $g_{\mathcal{T}} = 100$. With $\boldsymbol{g}_{\mathrm{q}}$ being a vector with the $n_{\boldsymbol{u}_{\mathbb{R}}}$-times repetition of g_{q} and $\boldsymbol{g}_{\mathcal{T}}$ being a vector with the $n_{\boldsymbol{u}_{\mathbb{Z}}}$-times repetition of $g_{\mathcal{T}}$, $\mathbf{G}$ can be constructed:

$$\mathbf{G}_{\mathrm{w}} = \operatorname{diag}([\boldsymbol{g}_{\mathrm{q}},\ \boldsymbol{g}_{\mathcal{T}}]). \tag{18}$$

The matrix $\mathbf{G}_{\mathrm{s}}$ for weighting the soft constraints of the voltage setpoints is also a diagonal matrix with equal entries g_{s} on the main diagonal. Here, g_{s} is selected so that $g_{\mathrm{s}} \geq g_{\tau} \geq g_{\mathrm{q}}$. This means that voltage problems are primarily solved by using reactive power. If this is not sufficient, the tap changer is used and lastly voltage setpoints deviations are permitted to allow convergence. In the investigated grid, $g_{\mathrm{s}} = 10^6$.

3 Study Case: Setup

3.1 Experimental Setup

The functionality of the proposed OFO controller is demonstrated simulatively. The controller is therefore not fed with real measurements as feedback, but virtual measurements from power flow calculations in the TG depicted in the left part of Fig. 2. The controller regulates the voltage magnitudes of the 220 kV/380 kV nodes of the TG/DG-transformers to a setpoint value or ensures compliance with a defined voltage band. The virtual nodal voltage magnitude measurements $\boldsymbol{y}_k$ at each instant k from the power flow calculation are fed to the controller at an interval of $\Delta t_{\mathrm{m}} = 1$ min. The controller then determines the control input $\boldsymbol{u}_k$ and transfers the signals to the actuators, i.e. to the DER or transformers.

3.2 Description of Investigated Grid

The grid analysed in this study case is a lightly meshed HV grid from the Simbench dataset. Details on the grid can be found in the publicly accessible dataset and its documentation [12]. The grid is a wind farm dominated 110 kV HV grid representing real HV grids from northern Germany and is located in a mixed semi-urban and rural region. In total, there are 64 stations and 95 lines (mainly over-head lines) with a total length of 1084 km. Three interconnection points to the overlaid TG exist. Fig. 2 shows the geographic single line representation of the grid. The grid is embedded in the TG model from the Simbench data set, which represents the German TG.

3.3 Load/Generation Profiles and Measurements

The Simbench data set contains time series for load and generation for all voltage levels. Low- and medium-voltage grids are integrated as PQ nodes by aggregating load and generation. The time series for the wind farms connected to the HV grid are based on representative probability distributions for wind

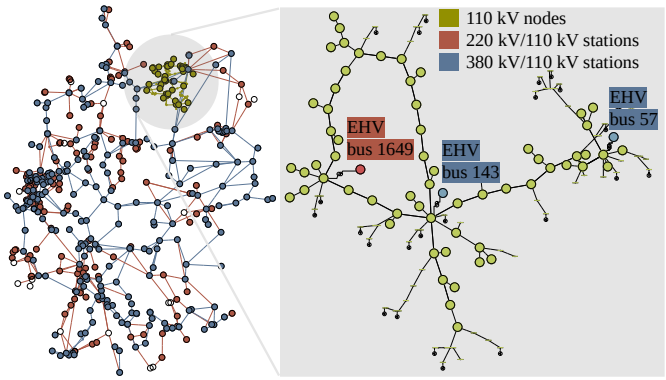

Fig. 2 Geographic single line representation of the Simbench TG and the investigated 110 kV HV grid.

speeds for the geographical location. Details about the modelling can be found in [12]. In general, the time series have a resolution of 15 minutes. However, the controller has a sampling rate of $\Delta t_{\mathrm{m}} = 1$ min. The load and generation time series are therefore linearly interpolated between the original time steps t, such that $t = (k - k \bmod 15)/15$:

$$\boldsymbol{p}_{1\mathrm{min},k} = \boldsymbol{p}_t - \boldsymbol{p}_{t+1} \frac{\boldsymbol{p}_{t+1} - \boldsymbol{p}_t}{k \bmod 15}, \tag{19}$$

$$\boldsymbol{q}_{1\mathrm{min},k} = \boldsymbol{q}_t - \boldsymbol{q}_{t+1} \frac{\boldsymbol{q}_{t+1} - \boldsymbol{q}_t}{k \bmod 15}. \tag{20}$$

3.4 Investigated time period

The period under investigation is a January day. The generation of the wind farms varies between about 0.25 and 1 pu in the period under consideration. The available reactive power reserves are sufficient to keep the voltage levels in the voltage band even without using the tap changer. If a voltage schedule is specified, it might be necessary to use the tap changer to find a solution. Without using OFO, the power flow results in voltage band violations at the TG nodes and in the DG.

4 Study Case: Results

4.1 Operation Within the Voltage Band

Initially, the controller is used to keep the voltage magnitudes in the HV grid and at the primary sides of the transformers within the permissible voltage band, i.e. $0.9\ \mathrm{pu} \leq |\boldsymbol{v}_k| \leq 1.1\ \mathrm{pu},\ \forall k \in \mathcal{K}$.

Fig. 3 shows the voltage magnitudes, the reactive power usage of DER and the active power losses. Starting at approximately 05:00, the voltage band at EHV bus 57 (green) is violated in the reference case without OFO. From 06:30, a violation also occurs at EHV bus 143 (red). Using the proposed controller, the voltage magnitudes remain within the voltage band at all times. Due to the loss minimisation objective of the OFO, the losses till around 05:30 are lower than in the uncontrolled case. Since the DER must be operated under-excited from about 05:30 in order to avoid voltage band violations the loss curves approach each other. As in the uncontrolled

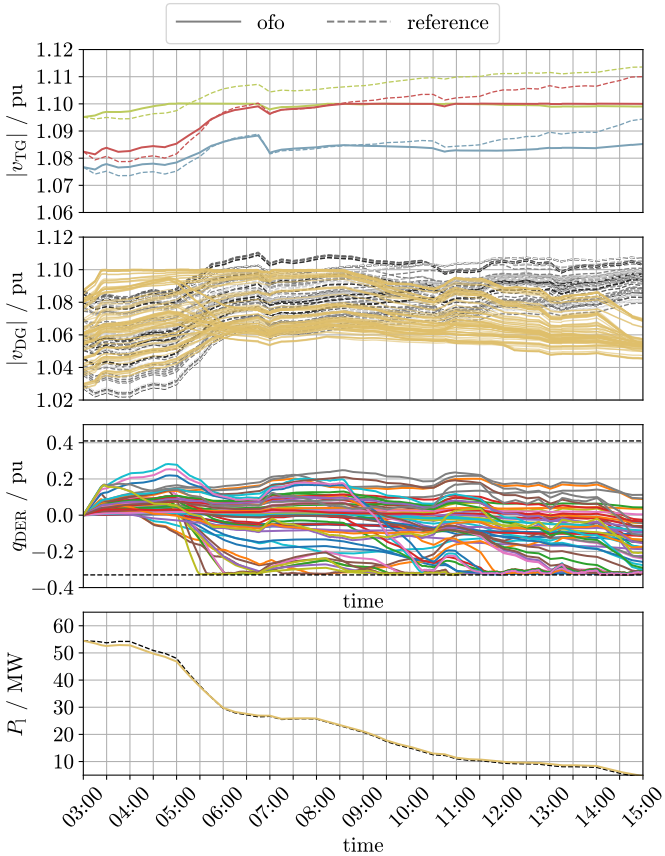

Fig. 3 From top: Voltage magnitudes at primary side of transformers, voltage magnitudes in HV grid, reactive power feed-in of DER and active power losses of HV grid. The dashed lines are showing the reference case, the solid lines show the results when using the OFO. The TG buses are: EHV bus 57 (green), EHV bus 143 (red), EHV bus 1649 (blue).

case voltage magnitudes are higher, current magnitudes are lower and therefore the losses are also lower, i.e. $\boldsymbol{p}_{\mathrm{l}} \propto |\underline{\boldsymbol{i}}|^2$. With regard to the reactive power, Fig. 3 shows an increasingly under-excited behaviour of multiple DER from 05:30, which is necessary to maintain the voltage band. The reactive power used can be influenced by g_q, which conversely has an impact on the step size in the direction of $\boldsymbol{\nabla f}$. However, due to the LOP, at least as much reactive power as needed to keep all voltage magnitudes inside of the voltage band needs to be provided. It should be noted that the tap changer is not utilised due to $g_{\mathcal{T}} \gg g_{\mathrm{q}}$ and sufficient reactive power reserves; thus, it is not shown here.

4.2 Compliance with Voltage Schedule

While compliance with the static voltage band was investigated in the previous section, compliance with voltage setpoints specified by a voltage schedule issued by the TSO is now to be investigated. The exemplary voltage schedule for all three TG nodes can be seen in Fig. 4. For the exemplary demonstration of the controller, it is assumed that all three nodes receive the same setpoint voltage specifications.

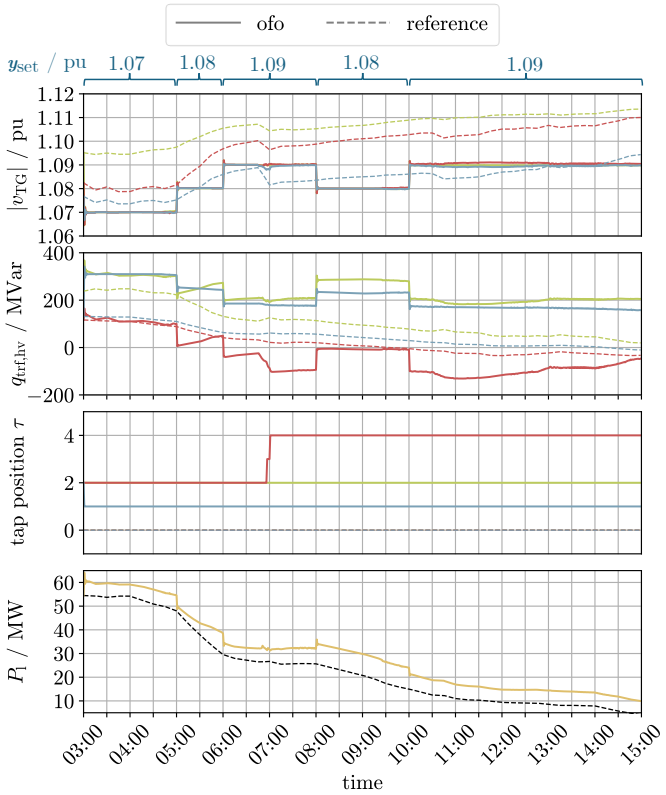

Fig. 4 From top: Voltage magnitudes, reactive power flow at hv side of TG/DG substation, tap positions and losses. Dashed lines indicate the reference case, solid lines the results using OFO. The buses are: EHV bus 57 (green), EHV bus 143 (red), EHV bus 1649 (blue). On top of the plot, the setpoints provided by the voltage schedule can be seen.

Fig. 4 shows that the controller manages to control the voltage magnitudes almost precisely to the setpoint. By modelling the voltage specifications as soft constraints, small deviations are possible, e.g. between 04:30 and 05:00, around 06:30 or at 10:30-13:00, but convergence is ensured. However, the control deviations are in the range of $\Delta v < 0.001$ pu and are therefore generally negligible for criteria such as static voltage stability. In contrast to Sec. 4.1, the tap changers are also used to maintain the voltage magnitude setpoints. As desired, the on-load tap-changer is only switched infrequently. The compliance with the setpoints leads to higher losses in the HV grid, which means additional costs for the DSO. On the other hand, compensation payments for the provision of ancillary services can be expected, e.g. in the Swissgrid concept for static voltage stability, see [13].

5 Conclusion

This paper demonstrates the potential of active distribution grids to support transmission grid voltage control through an online feedback optimisation (OFO) approach. By using a mixed-integer quadratic program in the controller, the proposed method uses reactive power of distributed energy resources in the distribution grid and tap changers to successfully control the transmission grid side of interconnection transformers. The robust performance of the controller, validated using the Simbench high-voltage benchmark grid, highlights its ability to adapt to varying operational conditions and ensure compliance with voltage schedules or defined voltage bands, while minimising grid losses, and adhering to operational constraints without requiring detailed knowledge of the grid state. This approach enables interconnection nodes to function as voltage-regulated PV nodes instead of PQ load nodes. Future research could focus on incorporating adaptive sensitivity estimation methods or the use of state estimation to determine the actual loss minimum in the control objective. Furthermore, future research should investigate how OFO-controlled distribution grids can enhance static voltage stability in transmission grids. The development of methods for calculating the time-dependent reactive power potentials at each interconnection node to provide them to transmission system operators is also a relevant topic.

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